

1. *Introduction*

2. *Background*

The first part of the paper discusses the background of the study.

3. *Methodology*

The methodology used in this study is a combination of qualitative and quantitative methods.

The data was collected through interviews, focus groups, and surveys.

The data was then analyzed using content analysis and statistical analysis.

The results of the study are presented in the following sections.

The first section discusses the findings of the interviews and focus groups.

The second section discusses the findings of the surveys.

The third section discusses the findings of the statistical analysis.

The fourth section discusses the findings of the content analysis.

The fifth section discusses the findings of the focus groups.

The sixth section discusses the findings of the interviews.

The seventh section discusses the findings of the surveys.

The eighth section discusses the findings of the statistical analysis.

The ninth section discusses the findings of the content analysis.

The tenth section discusses the findings of the focus groups.

The eleventh section discusses the findings of the interviews.

The twelfth section discusses the findings of the surveys.

The thirteenth section discusses the findings of the statistical analysis.

The fourteenth section discusses the findings of the content analysis.

The fifteenth section discusses the findings of the focus groups.

The sixteenth section discusses the findings of the interviews.

The seventeenth section discusses the findings of the surveys.

The eighteenth section discusses the findings of the statistical analysis.

The nineteenth section discusses the findings of the content analysis.

The twentieth section discusses the findings of the focus groups.

The twenty-first section discusses the findings of the interviews.

The twenty-second section discusses the findings of the surveys.

The twenty-third section discusses the findings of the statistical analysis.

The twenty-fourth section discusses the findings of the content analysis.

The twenty-fifth section discusses the findings of the focus groups.

The twenty-sixth section discusses the findings of the interviews.

The twenty-seventh section discusses the findings of the surveys.

The twenty-eighth section discusses the findings of the statistical analysis.

The twenty-ninth section discusses the findings of the content analysis.

The thirtieth section discusses the findings of the focus groups.

The thirty-first section discusses the findings of the interviews.

The thirty-second section discusses the findings of the surveys.

The thirty-third section discusses the findings of the statistical analysis.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Tech. Examination (EE)

April 2013

EE407 Commissioning of Electrical Equipments

Date: 23.04.2013, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Enumerate the points to be taken into consideration at the time of preparing the schedule for maintenance. [04]
- (b) Explain following terms [03]
- (1) Polarization index of transformer
 - (2) Degree of protection
 - (3) Single phasing of three phase induction motor.
- Q - 2 (a) Explain impulse test on power transformer with neat sketch of test setup. [07]
- (b) Which are the conditions for parallel operation of transformers? Explain load sharing problem with parallel operation of transformer in brief. [07]

OR

- Q - 2 (a) Give causes, consequences and remedies for the following abnormal conditions of power transformer (i) over fluxing (ii) excessive core heating (iii) moisture in oil and (iv) internal short circuit. [08]
- (b) Explain drying out process of transformer. [06]
- Q - 3 (a) Explain back to back test of DC machine with circuit diagram. [07]
- (b) Explain following abnormal condition of induction motor (i) crawling (ii) pullout torque and stalling. [07]

OR

- Q - 3 (a) Explain heat run test of induction motor with circuit diagram. [07]
- (b) Discuss commutation problems of DC generator and factors affecting speed of DC motor. [07]

SECTION – II

- Q - 4 (a) Answer the following questions. [07]
- (1) Draw the connection diagram of earthing transformer when secondary is star

connected showing current distribution in case of single line to ground fault.

- (2) Give the equation to find out the soil resistivity with the use of four electrode method.
- (3) What are the auxiliaries required for tripping test of an alternator?
- (4) State advantage and disadvantage of EHV alternator.
- (5) State various type tests to be conducted on lightning arrester.
- (6) Define transfer potential.
- (7) Explain how CTC type equipment used for fire precautions in substation?

- Q - 5 (a) Why resistance earthing is adopted? Explain resistance earthing with necessary diagrams, advantages and disadvantages. [07]
- (b) State the various commissioning tests on circuit breaker. Explain any two of them in brief. [07]

OR

- Q - 5 (a) Explain the design of earthing grid in context with (i) data needed for the design and (ii) design steps for earthing system. [07]
- (b) Enumerate various points to be considered at the time of installation of overhead lines. [07]
- Q - 6 (a) Which is the test to be conducted on underground cable to locate the fault distance in case of earth fault? Explain the same with necessary diagram. [07]
- (b) Which are the high voltage tests to be conducted on alternator? Explain each of them in brief. [07]

OR

- Q - 6 (a) Give causes and remedies for following condition of alternator (i) heating of copper conductors (ii) heating of brushes and slip ring (iii) bearing troubles. [07]
- (b) Describe following terms related to alternator in brief (i) leakage reactance (ii) transient torques (iii) insulation resistance measurement. [07]
